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Making Theory Come Alive through Practice-based Design Research

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Abstract

The overall aim of this paper is to demonstrate how practice-based design research is able not only to challenge, but also push toward further development of some of the background assumptions of emotion theories as used within design research. In so doing, we wish to increase knowledge on a central epistemological question for design research, namely how practice-based design research can be a vehicle for the construction of new theory for design research.

To set the scene, we start out by providing a brief literature survey, where we discuss insufficiencies of some previous treatments of our subject matter. Secondly, and in order to remedy these insufficiencies, we introduce "dynamic research sketching" as a new explanatory tool for understanding how design research can be a source for generating new theory. Further, we apply dynamic research sketching to a research project thereby explaining how the design of a computer game has called for a rethinking of basic notions within emotion theories.

Keywords
Practice-based design research, dynamic research sketching; design experiments

Introduction

The overall aim of this paper is to demonstrate how practice-based design research is able not only to challenge, but also push toward further development of some of the background assumptions of emotion theories as used within design research. In so doing, we wish to increase knowledge on a central epistemological question for design research, namely how practice-based design research can be a vehicle for the construction of new theory for design research.

Over the last decade or so, many attempts have been made in order to shed light on the role of theory in practice-based design research. The most significant contributions have come from general design studies (Friedman, 2003) or interaction design research (Fallman, 2003, 2005, 2008; Brandt & Binder, 2007; Haynes & Caroll, 2008; Redström, 2010; Zimmerman & Forlizzi, 2008; Zimmerman et al. 2010). Roughly speaking, these contributions deal primarily with two research questions: (1) What is the nature of theories used in design research? (2) In what ways can theory inform practice-based design research? While these questions are no doubt of fundamental value, we argue that there is an equally important need to address a

third question: (3) How is practice-based design research able to feed back into and transform theory?

This third question, which has only received scant attention so far in design research, defines the crux of the problem that we want to address in this paper. To set the scene, we start out by providing a brief literature survey, where we discuss insufficiencies of some previous treatments of our subject matter. Secondly, and to remedy these insufficiencies, we introduce "dynamic research sketching" as a new framework for understanding how design research can be a source for generating new theory. Moreover, we apply dynamic research sketching to a research project thereby explaining how the design of a computer game has called for a rethinking of basic notions within emotion theories. Finally, we discuss how knowledge production in practice-based design research differs from mainstream lines of research.

On the relation between theory and practice in design research

Much attention has been paid already to the role of theory in design research. Existing literature on the topic can be roughly categorized into three strands of work that we describe as the *essentialist, instrumentalist* and *constructivist* strand. These strands differ from each other in as much as they offer different methods for working with theory in practice-based design research.

To the essentialists, a theory is a coherent system of concepts that is *applied* in design research to increase understanding of a given situation. According to the instrumentalists, on the other hand, a theory is more like a toolbox full of concepts that can be used to *inform* the design and making of an artefact. And the constructivist strand is characterized by a focus not on how theory may apply to design research or inform design practice, but the other way around: how design practice can be a vehicle for *transforming* theory. In this section we select and review research literature thereby providing a more detailed description of each of these strands. The overall purpose of this review is to provide a better understanding of how theory and practice are able to integrate in design research.

The essentialist strand

A proponent of the essentialists strand is Friedman (2003), who lays out a comprehensive framework for understanding the nature of theory and how different kinds of theory are at work in design research. Friedman's conception of theory and its use in design research are heavily influenced by scientific ideals and definitions. In science, a theory is primarily thought of as "providing principles of analysis and explanation of a given subject matter" (Friedman, p. 514). By applying a theory in a research project, researchers strive to secure a *systematic* method of inquiry, which is the hallmark of scientific research.

Insofar as he heralds these scientific standards, Friedman ends up giving the impression that design research is most of all about studying and explaining various kinds of subject matters relevant for design research. Theory, then, is what a design researcher applies to make his or her method of inquiry systematic, for instance as a way of grouping observations or data under general categories. This is what – according to Friedman – qualifies design research as a scientific discipline and which makes it different from design practice. In fact, Friedman is reluctant to attribute any constitutive role to practice, because, in his view, "it is not practice, but systematic and methodological inquiry into practice that constitute design research" (Friedman, 512). By forfeiting this view, Friedman does not acknowledge the making of artefacts and designerly experiments as legitimate research methods. So, while many insights about the nature of theory can be gained from Friedman's study, he generally fails to explain what role practice can play in the construction of new theory.

The instrumentalist strand

Instrumentalists however provide a more elaborate account of how theory may integrate with practice in design research. Generally, instrumentalists share the view that a theory offers itself as a toolbox full of concepts that can be used to inform the design of an artefact. Such a view is defended by Caroll (1987), Caroll & Kellogg (1989), Barnard (1991), and most recently by Haynes & Caroll (2008). In contrast to Friedman, Haynes & Caroll (2008) do not conceive of theory as something that is merely applied to the study of practice, but as a guiding philosophy that help designers make decisions concerning the shaping and form of an artefact. In this way, they say, the artefact may "embody" a theory.

Even though the instrumentalist account see theory as an integrated part of practice, we argue that it is still too limited insofar as it depicts the theory-practice nexus as a one-way information process. It is as if theory for Haynes & Caroll (2008) comes to design researchers as a pre-existing package of concepts and ideas that are fed into design practice, while the mutual exchange between practice and theory remains unexplored. This is reflected in their analysis of the by now classic example of the Xerox Star workstation, which they claim, "was the first information processing system to fully embody the theory of direct manipulation human-computer interfaces" (Haynes & Caroll, p. 166). However, the theory of direct manipulation did not exist as a fully-fledged theory in 1981, when Xerox Star was introduced. Quite, on the contrary, it did not emerge until twenty years after direct manipulation interfaces began to appear (Zimmerman et al, 2010, p. 312).

The theory was developed by synthesizing a set of visionary ideas introduced throughout the late 1970s and the beginning of the 1980s by people like David Liddle, Stu Card, Ben Schneiderman and others, and as a result of recurrent research experiments with new digital technology available at the time at Xerox Parc. For instance, bitmapped displaying technologies which allowed users to directly manipulate digital processes in a computer by clicking on graphical icons on a screen instead of having to learn the complex programming language of command-line interfaces. Hence, what the instrumentalist account suffers from is lack of knowledge as to how recurrent experimentation in design research contributes to theory construction (other critical voices against the background assumptions underlying the instrumentalist strand can be found in Landauer, 1991).

The constructivist strand

This question has become of central interest to researchers belonging to the constructivist strand, who generally adhere to Frayling's (1993) original idea of *research-through-design*, i.e. that processes and experiments from design practice should be recognized as legitimate methods of inquiry. Among these are Zimmerman et al. (2010, p. 311) who argue that the process of designing and making an artefact can form the basis for "producing design theory that is distinctly different than scientific theory" (for similar views, see e.g. Fallman, 2003, 2005, 2008). Furthermore, Zimmerman and Forlizzi (2008) suggest that a distinction should be made between two different methodological approaches driving theory construction within research-through-design: (i) a *philosophical* approach, where researchers wish to investigate and extend "a previously articulated theory through a process of making" (e.g. the theory of 'ludic interaction', 'rich interaction', 'emotional interaction', etc.); and (ii) a *grounded* approach, where theory grows out of researchers focus "on real-world problems by making things that force a concrete framing of the problem." (Zimmerman & Forlizzi, p. 43)

However, we argue that the whole idea of research projects being categorized into one out of these two methodological approaches rests upon a false dichotomy. More often than not the philosophical and grounded approaches are co-present in a research-through-design project. For instance, in the design research project we will be discussing in the next section it was investigated how emotion theories were able

to inform the design of a computer game for hospitalized children. But, at the same time the design of the computer game was framed by a real-world problem, namely the need for hospitalized children to communicate how they feel about their treatment to nurses and doctors at the hospital. What is important here is not so much that this project demonstrates the insufficiency of Zimmerman and Forlizzi's distinction, but rather to see that it is precisely through the interweaving of the philosophical and grounded approach that new theory construction took place. More specifically, the outcome of the research project called for a significant rethinking and extension of emotion theories.

Under closer scrutiny, it turns out that this is not an isolated incident, but a common characteristic of research-through-design leading to theory transformation. Our main argument is therefore that theory transformation should be conceived of as relying on the dialectics between a real-world problem, recurrent experimentation and design theory. In fact, we claim that further insight into the role of experiments holds the key for understanding how the processes of designing and making artefacts produce theory. Zimmerman et al. (2010) correctly point out that there is no agreed upon method to document knowledge that emerge through this type of design research, but they are unable to deliver this method, because the notion of experiment is absent from their account.

However, Brandt and Binder (2007) lay out what, in our view, must be considered a more valuable framework for understanding experiment-driven design research. According to this framework, a design research project can be understood in terms of three basic elements: a *research question* (Q) guiding the inquiry by setting an overall scope; a *program* that specifies "an area of exploration setting goals of what is to be achieved by the design"; and experiments which explores the program and serve as a vehicle for knowledge production. Brandt and Binder suggest the following diagram as a way of depicting how a research project can be explained with the use of these categories:

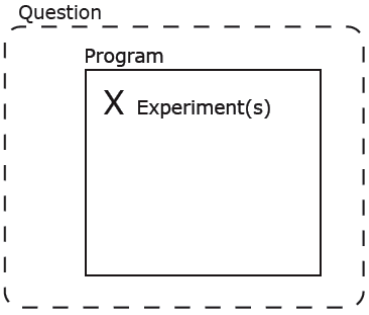


Fig. 0. Diagram by Brandt & Binder

We agree with Brandt and Binder (2007, p. 4) that this diagram is of much use as it allows for a "methodological grounding of design research driven by designerly experiments". In so doing, it provides means for documenting how knowledge emerges from design research. Moreover, it does not instigate an artificial divide between methodological approaches that usually run in parallel in a research project. For instance, the question as to how real-world problem may force a framing of a research project is represented by the program, which defines a design space, end users, context of use, economical constraints, technological specifications, and so on. Likewise, Zimmerman & Forlizzi's suggestion that a question formulated out of previously articulated theory can be set to drive the research project can easily be depicted by the research question in the diagram.

Yet, the diagram suffers from two central limitations that need to be addressed. There is an inconsistency between the diagram, which gives only a static view of a project, and Brandt and Binder's (2007) notion of experiment-driven design research, as a process of on-going interactions between designerly experiments, program and research question. In this way the diagram is unable to explain, for instance, how a research question gets modified through experimentation, or how outcomes of experimentation call for new theory formation.

Secondly, theory does not figure in the diagram. Does theory help in articulating research questions or to refine programmatic considerations? Only insofar as the diagram is made dynamic and extended with a visual element for theory, will it be able to answer such questions.

Our main contribution falls under the constructivist strand. We introduce dynamic research sketching as a way of making Brandt and Binder's diagram capable of visualizing how theory construction emerge out of a dialectics between research question, program and designerly experiment (cf. Redström, 2011).

Dynamic Research Sketching

In this section we use an on-going research project for exemplifying the explanatory strength of dynamic research sketching as compared to Brandt & Binder's diagram. The aim of this project is to develop a computer game for hospitalized children – called the Child Patient game (CP game) – that can map children's subjective feelings, through their interactions with narrative structure and game play (Knutz & Markusen, 2010). Basically, the CP game allows hospitalized children to attach emotions and "secret powers" to a fictional character, trying to cope with a hospital-situation similar to that of their own.

The intention of designing this game play is to investigate an initial research question: Is it possible to gain information about the emotional lives of paediatric patients through gaming (RQ1)? However, the process of designing the computer game itself generated a new research questions: What is the relation between fictional emotions chosen in the game and the children's own felt emotions during hospitalization (RQ2)?

This research question occurred as an outcome of a research-through-design approach that involved the set up of two different experiments and they were important in two regards. Not only did they raise an awareness of knowledge gaps inherent the design theories initially informing the project. They also forced theoretical considerations - and the mode of "asking" and "thinking" and "doing" - throughout the project to change, modify, and drift.

Before going deeper into a discussion of the process of theory transformation, it is useful to sketch the generation of research question as a result from the interaction between Research Questions, Program and Experiment. Making a first small adjustment to Brandt & Binder's diagram enable us to sketch this:

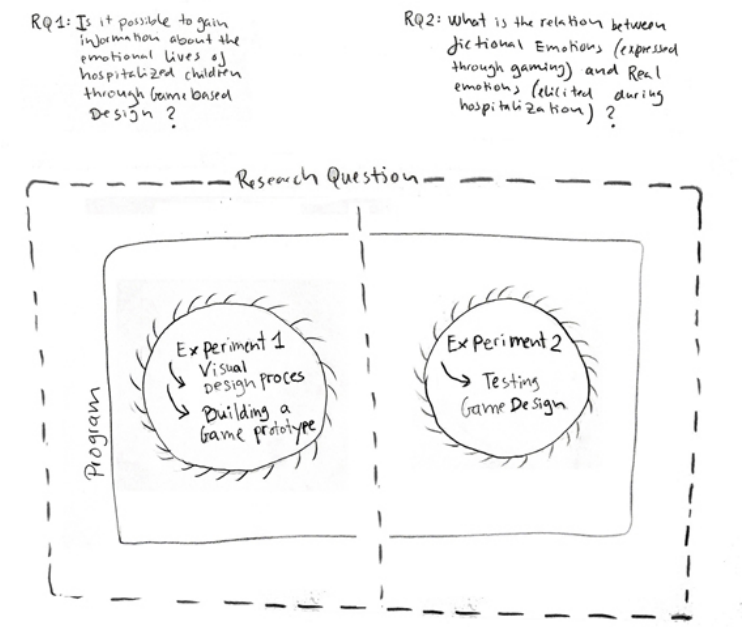


Fig. 1: Sketch 1

Sketch 1 shows that this research project has been undergoing two Experiments within a Program: First of all a computer game has been designed to allow hospitalized children to inform staff and researchers about their emotional experience (Experiment 1). Secondly, the computer game has been tested with a group of patients and non-patients, using the game as an alternative method of inquiry (Experiment 2).

Experiment 1 and 2 are framed by the same Program (that contextualizes the experiments). In this case the program defines a specific context at a hospital (children ward), a target group of hospitalized children (age 4-6) as well as staff and parents, who all have their daily routines at the ward. Further, Sketch 1 shows that the project was initiated with RQ1: Is it possible to gain information about the emotional lives of hospitalized children through gaming? But Sketch 1 doesn't say anything about the method used or how the project is linked to other research fields or theories relevant for the project. This is what Brandt and Binder refer to as the theoretical "genealogy" of the project. The genealogy is understood much as a theoretical umbilical cord that nurtures the project with precise concepts that allows for making exact conceptual distinctions in the understanding of a subject matter.

Insofar as the genealogy and the whole question as to how theory enters the project is not represented by Brandt & Binder's diagram (Fig. 0), the sketch cannot visually explain the dialectics underlying new theory formation. To make up for this shortcoming we included genealogy into a new research sketch that aims to capture how the generation of questions co-evolve with needs for integrating new theories into the project. Moreover, to gain further insight into the constructive role of the experiments in this process, we decided to use the sketch as a kind of magnifying glass providing a microscopic look into the dialectics between experiment and theory.

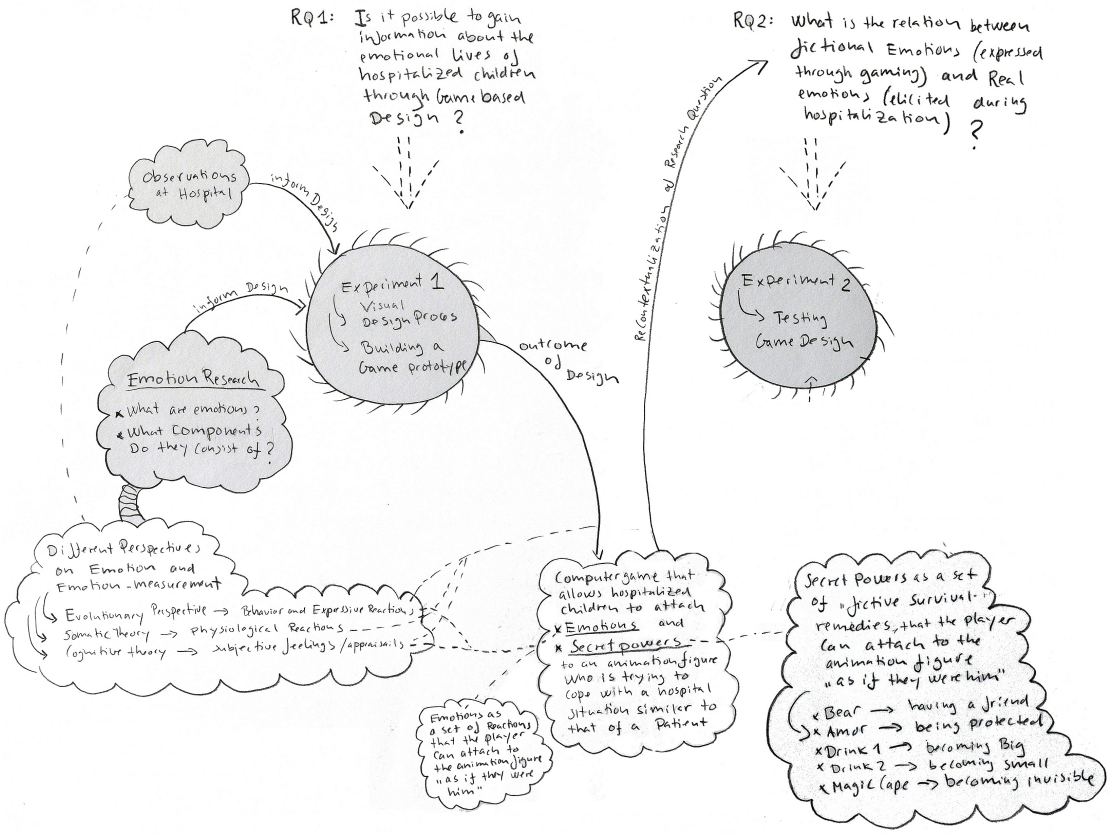


Fig 2: Sketch 2

One can see that Experiment 1 is clearly shaped by emotion theories as found in psychology, cognitive science and philosophy as well as by emotion-driven approaches to design (see e.g. Desmet, 2002, 2008; McDonagh, Hekkert, van Erp, & Gyi, 2004; Norman, 2004; Norman & Ortony, 2006). In the initial phase, these theories and approaches was used to work out a fine-grained understanding of what an emotion is, how emotions are elicited and the way in which emotions are typically expressed either through facial expressions or bodily reactions. In particular three theories of emotions were important for the project:

- *The Evolutionary Theory,* which was originally founded by Darwin, sees emotions as behaviours, expressive reactions and readiness to respond. Emotions within this perspective serve a functional purpose for survival: They help “organisms to deal with key survival issues posed by its environment”.

- *The Somatic Theory of Emotion*, that leads back to the founders of modern psychology, notably William James and Carl Lange, and which has received renewed interest from researchers such as Damasio (1994) and Prinz (2004). According to this theory the experience of an emotion is a direct result of a sensed change in bodily states. This change simply is the emotion: "We do not cry because we are sad, we are sad because we cry" as James is famous for having proclaimed.

- *The Cognitive Theory of Emotion*, which focus on how emotions are generated by the immediate judgments we make about the world. Magda Arnold, who is one of the pioneers of the cognitive approach, claimed that the appraisal (the valuation, the direct judgment) is the core of all emotions. She argues that emotions should be understood as "the felt tendency toward anything intuitively appraised as good (beneficial) or away from anything

intuitively appraised as bad (harmful)". It is not the event itself, but the meaning that we attach to this event, which is responsible for our emotion.

Insights from all of these perspectives were brought into the design of the CP game so that it allows children to attach a number of so-called basic emotions to an animation figure, who then changes (change of body state) and express emotion (through behaviour or facial expressions) according to the chosen emotion (the judgement of the children). However, the game does not simply embody these emotion theories, as Haynes & Caroll would have it. On the contrary, in carrying out the experiments involved in designing the game (drawing, storyboarding, animating, prototyping, programming, etc.), the theories were challenged and a need for broadening the theoretical grounding of the project became clear.

This has to do with the fact that Experiment 1 did not *only* produce a game that allowed children to attach emotions to an animation figure, it *also* produced a game world of its own; a narrative that allowed imaginary experiences to take place. These experiences are referred to as "secret powers".

Secret powers are a set of fictive survival tactics that the player can attach to the animation figure, which is trying to cope with a hospital situation similar to that of a patient. Secret powers can be a magic cape that makes the animation figure invisible, a drink that turns him into a grown-up or an iron amour that will protect him and make him invulnerable. These secret powers are (like the emotions in the game) a set of reactions that the player can attach to the animation figure "as if they were him".

Several questions occurred during the visual design process. Questions that emotion theories were unable to answer: Do imaginary experiences such as possessing secret powers have an effect on real-world experiences? How can we account for the relation between the player experience (that of a child patient) and that of a fictive character (who is also a patient in the game world)? What is the relation between fictional emotion expressed through gaming and real emotions elicited during hospitalization?

All of these questions called for a substantial broadening of the existing theoretical perspectives on emotions, but only the last one was selected as a new research question (RQ2) that we wanted to explore further by way of Experiment 2. In so doing, it became clear that the project needed a new genealogy: narrative theory. This is visualized in Sketch 3:

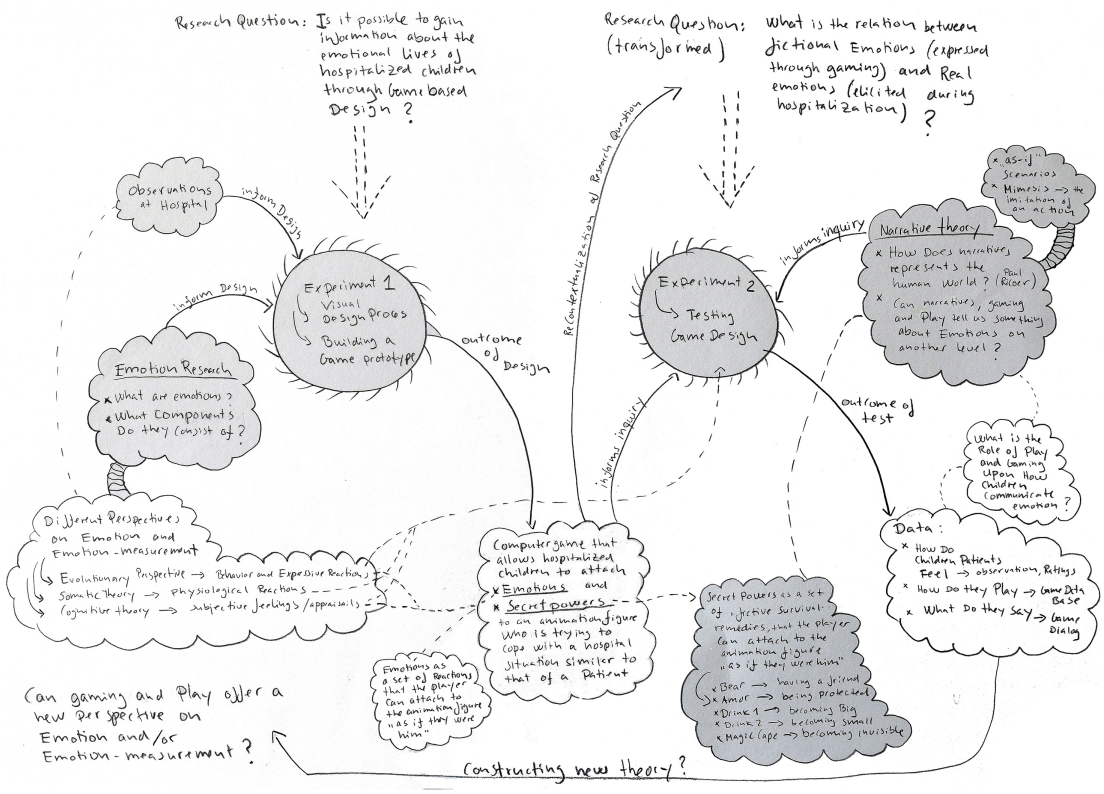


Fig 3: Sketch 3

What narrative theory has to offer, that emotion theories are lacking, is above all the concept of fictional emotions. Generally, emotion theories deal only with peoples felt emotions as in subjective feelings, behavioural expressions, and somatic reactions, not with fictional emotions played out in narrative as-if scenarios.

Within emotion research there is general agreement that emotion implicates Feelings, Actions and Thought (Prinz, 2004). This means that emotions entail a complex interplay between bodily responses as well as rational and cognitive processes. Fictional emotion also implicates Feelings, Actions and Thought, but the stimuli are to be found within the fictional world: we can cry when we experience a sad movie wherein the character, we sympathise with, will die. Or we can identify with a game character and act "as if we were him". These experiences belong to the game of make-believe (Kendall, 1990), which might or might not be different from the emotions exposed during a real situation. Real emotions as well as fictional emotion are acute and object oriented. But the exact relation between these two is not clear.

By informing Experiment 2 with narrative theory, the project aims to shed light on a "blind spot" within emotion theories and illuminate how the players felt emotions (during hospitalization) relate to the players imaginative experiences of a character's emotion (during the interaction with the computer game).

Theory Push – inside view

With a new genealogy (Narrative Theory) attached to Experiment 2, the "mode of asking" and the theoretical approach toward the testing of the CPgame needs modification. This is done by letting the CPgame function as an alternative method of inquiry. An inquiry that enables researchers and designers to interpret and make sense of the emotional experiences communicated by children though gaming.
In Sketch 3, this process is visualized: Experiment 1 raises a re-articulated research question and lets Experiment 2 drift towards an alternative inquiry, that will produce data based on:

- How children FEEL, when being examined in Hospital (Real Emotions)
  (Method: observations and visuel ratings about emotions, from staff and parents)

- How children PLAY (Fictional Emotions)
  (Method: designing a database for the game, that records play-pattern)

- What children SAY, about the fictive character (Fictional Emotions)
  (Method: Game dialogs with children while playing the CPgame)

This data points into the theoretical gap (within emotion theories) by addressing fictional emotions. In doing so Experiment 2 calls for a substantial broadening of the existing theoretical perspectives on emotions, by adding a (new) narrative perspective to the theories of emotions and hereby constructing theory. This is the Theory Push, visualized through research sketching.

Based on the processes of research sketching (Sketch, 1, 2 and 3) we are now able to draw a schematic diagram that captures the dialectics underlying theory construction in practice-based research:

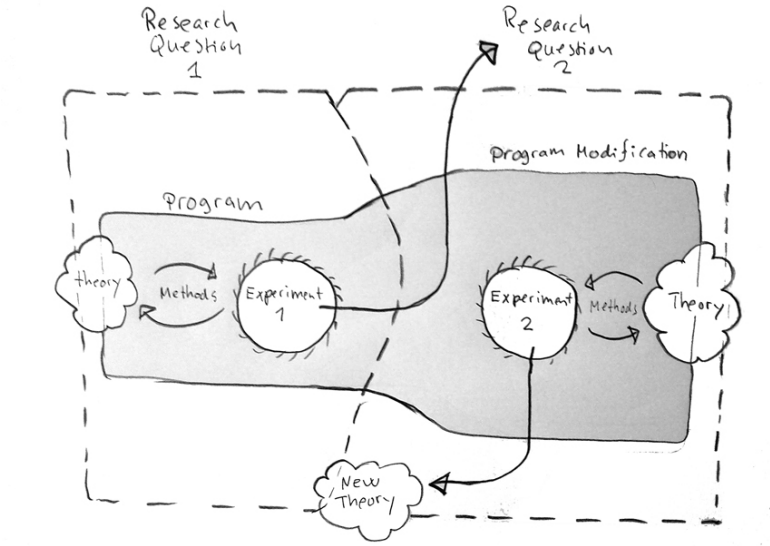


Fig.4: Diagram for "Dynamic Research Sketching" (Diagram DRS)

The Diagram DRS can explain (on an abstract level) - how the design of a computer game may push towards integrating the concept of “fictional emotions” into the theoretical understanding of how emotion research can play a role in design.

Theory push – an outside view

The role of experiments in science is long and profound. In many lines of research, experiments – in the field or in the laboratory – have a crucial role for the conduct of science.

Experiments played a crucial role in Galileos rejection of Aristotle’s law of gravity. Also the works by for example Newton and Einstein were based on experimental approaches. In this way experiments can be seen as tools for pushing theory.

However, in mainstream lines of research today, experiments take another, often more limited role. They are carried out with the goal of verifying, falsifying, or establishing the validity of a hypothesis. The experiment can thus be seen as a method of testing - with the goal of explaining – our images of reality.

In research through design, the experiments take slightly different positions. Partly because the experiment is part of design practice, where experiments predominantly are used to establish images of future realities or opportunities in contrast to present realities. Partly because the experiment is a tool in the research process. In this way the experiment has a key position, since it facilitates a meeting of practical, often tacit, knowledge with academic, most often codified knowledge.

As is revealed in the CPgame example above, the experiments have a double function vis-à-vis the design program forming the core of the research project and the generation of theoretical insights.

Though the experiments, changes in the program, the research question as well as the theoretical framework is brought about. On the other hand, emerging changes in the research question and the theory construct will also lead to a quest for molding the experimental agenda.

Thus, the aim of the design program has changed from a sole focus on how emotions of hospitalized children can be traced and made explicate by help of the CPgame, the aim – and program - has changed to an additional focus on fictional emotions versus real emotions. In this way the insights from the experiments have changed the boundaries of the program (cf. Fig. 4).

However, the insights gained through the experiments have also regenerated the theoretical framework of the study, in this case by the inclusion of narrative theories and propositions, which in turn have enriched the emotional theory foundation informing the program in the first place.

Finally, the research question set out has changed, since the insights from the experiments have suggested that the original RQ should drift (is it possible to gain information about the emotional lives of hospitalized children though game-based design?).

On top of this the theoretical knowledge gained through the design driven experiments is the notion of fictional emotions, leading to a conceptual blend enriching emotion theory as well as research on narratives.

Research-through-design in perspective

Bringing the research presented in perspective, we may conclude that design research – as research through design - challenge and has the potential to enrich knowledge production in mainstream research alleys in several ways:

- While mainstream research, including case-based research, tend to let the research question guide the journey of knowledge creation in a goal directed and thus predetermined way, research through design, as exemplified here, allows the research question to float with insights gained under way.
  This has connotations to what Quinn (1980) labels ‘logical incrementalism’. It is seen as an approach in which many small (often unplanned) incremental changes are added as new insights or changes as the context calls for it. In view of this perspective, the research process is seen as an incremental process of design experiments and insights, leading to a fluid, less static, but controllable research process.
- While mainstream research tends to find the research process predictable, research through design find that the research process cannot be predicted. The research process come to life through an action-based research approach.
- While mainstream research emphasizes aim and program as the framework for the research process to come, the program – in design driven research - is allowed to be molded by the knowledge - experiences and insights - generated by the experiments and the theory expansion process

- While mainstream research mainly uses experiments to validate theoretical propositions, research through design turns the coin up-side-down and use the experiment to mold and further develop research questions and theoretical propositions.
- While mainstream research tends to see academic knowledge as the only valid knowledge, research through design finds that conceptual knowledge generated by the design experiment produce insights and knowledge that may inspire and enrich academic knowledge.

This, of course, leaves the reader with an image of an unguided research process in which some of the established key claims to and restrictions on research are disregarded. Whether this will be the case, relies very much on the intuitive capability of the researcher and the ability to document the research process. Saying this, it is also important to stress that theorizing is often undermined by methodological restrictions in favor of validation more than practical relevance. In this way they tend to disregard creative approaches as imagination, experiments, representation (Weich, 1989).

4. Conclusion

In this paper we have tried to increase knowledge as to how practice-based design research can be a source for new theory formation. The purpose of our initial literature survey was to critically examine some of the most dominating ideas concerning the theory-practice nexus in design research. Further, we wanted to unravel what we see as being the basic principle of theory construction, namely the dialectics between recurrent experimentation, program and research question. While the most useful understanding of this process can be found within the constructivist strand, the diagrams and frameworks offered by constructivists are too limited and require further development.

We have introduced dynamic research sketching as a new explanatory tool for visually grasping the dialectics underlying new theory construction in design research. In so doing, we remedy the inherent shortcomings and inconsistencies of the existing frameworks and diagram, notably Zimmerman and Forlizzi (2009) and Brandt and Binder (2007).

Admittedly, our diagram of dynamic research sketching rests upon the application to a single case example. Yet, the diagram we lay out is in no way claimed to be exhaustive. Nonetheless, we hope that dynamic research sketching can become a valuable tool for explaining various forms and principles of theory construction, for instance, theory extension, theory refinement and theory renewal. Such an endeavor has though to await future work. The aim of introducing dynamic research sketching in the present paper was far more modest, namely, to show that far from killing theory, practice-based research actually holds the key for making theory come alive.

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